Enhancing Inclusive Education in Sri Lanka through a Personalized Multimodal Learning Platform for Children with Neurodiversity

25-26J-218

ADHD، Down–Syndrome and Autism

SLIIT
FACULTY OF COMPUTING

Our Team

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Research Objectives

Main Objectives

A personalized, gamified learning system that empowers neurodiverse learners (with Down syndrome, ASD, ADHD) to master life skills and job roles for independent living and employability.

Sub Objectives

- Design and develop a personalized learning system
- Implement multilingual support with speech feedback
- Develop progress tracking and parental feedback system
- Ensure accessibility and user-friendly UI/UX for neurodiverse students

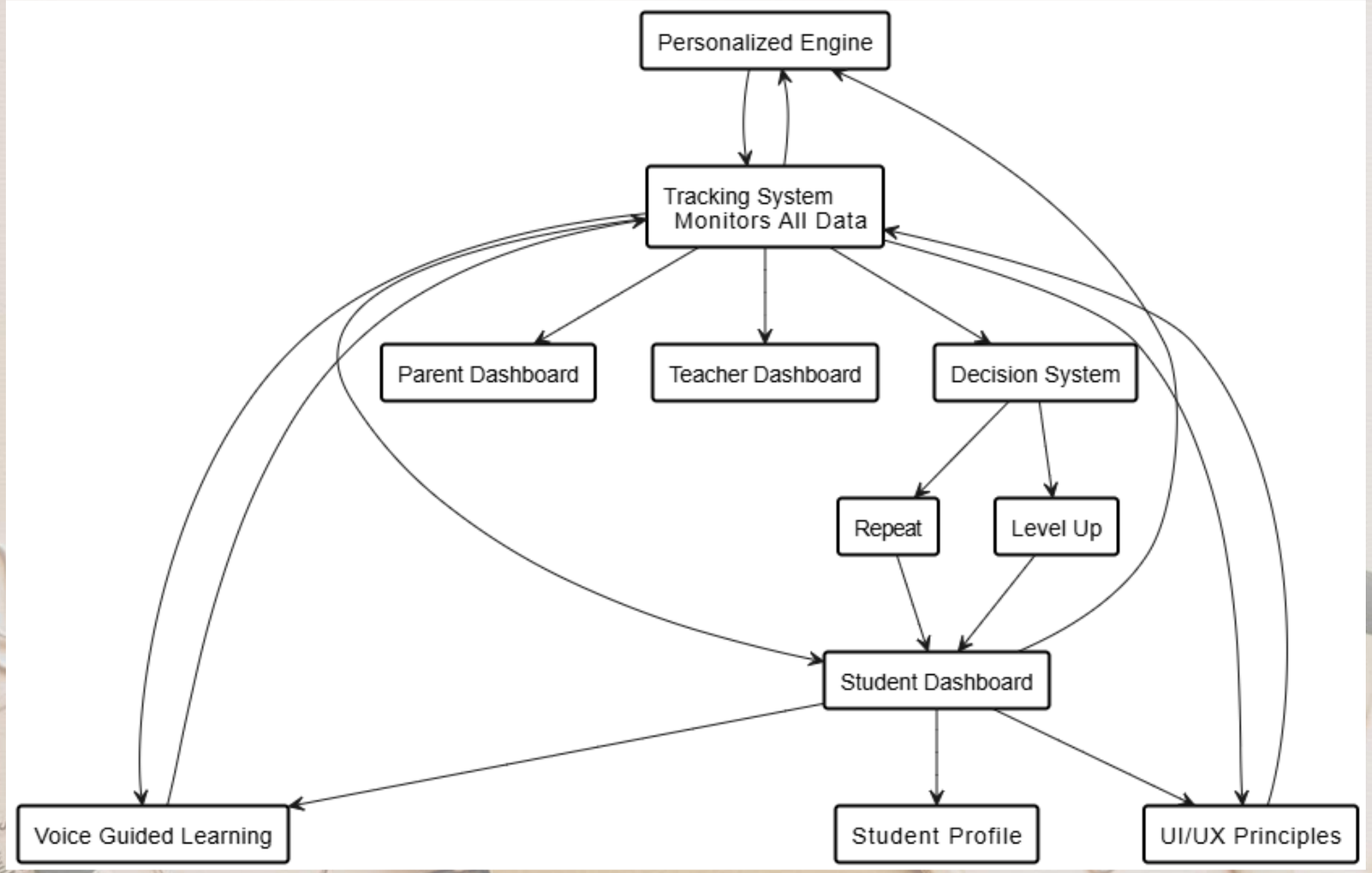
Introduction

The project aims to develop an e-learning platform to support children with learning disabilities such as Down Syndrome, ADHD and ASD.

Modules :

- Life Skills Module
- Job Role Simulation Module
- Communication and Social Skills Module
- Behavior & Emotional Regulation Module

System Diagram



IT22341518 - Fonseka W D C

Specialization : Information Technology

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Personalization Engine

Knowledge Gap / Problem Definition

- Current game-based learning and AAC apps provide customization, not personalization
- Limited ability to predict learning needs

Proposed Creative Solution – Personalization Engine

- Real-time adaptation
- Dynamic Feedback & Motivation
- Support for Neurodiverse & Multilingual Learners

Personalization Engine

Shortcomings of Existing Systems

App / Feature	

Personalization Engine

Key Domains of specialization-based Knowledge utilized

- Artificial Intelligence & Machine Learning
- Data Science & Analytics.
- Educational Technology.
- Software Engineering (MERN + Supabase)

Specific Latest Technologies utilized

Backend AI/ML:

- TensorFlow.js

Database & Tracking: Supabase (Real-time event tracking).
Engine Orchestration: Node.js (Express)
Integration with Frontend: React Native

Personalization Engine

Data Availability and Ethical Clearance

Data Required:

- Time taken per activity/lesson.
- Accuracy of answers
- Time spent on a screen/activity
- Drop-off points
- Frequency of interaction (taps, clicks, voice responses).

Availability:

- Gathere from the Datacener in Ayati Center, Ragama

Ethical Considerations:

- Store only anonymized learner IDs in Supabase.
- No sensitive personal/medical data beyond what is needed for personalization.
- Local ethical clearance from schools/institutes working with special-needs children.

Personalization Engine

How to Evaluate

MJF Charitable Foundation

www.mjffoundation.org

We plan to refine and adjust the feature based on insights gathered from the pilot survey conducted with students during the development stage.

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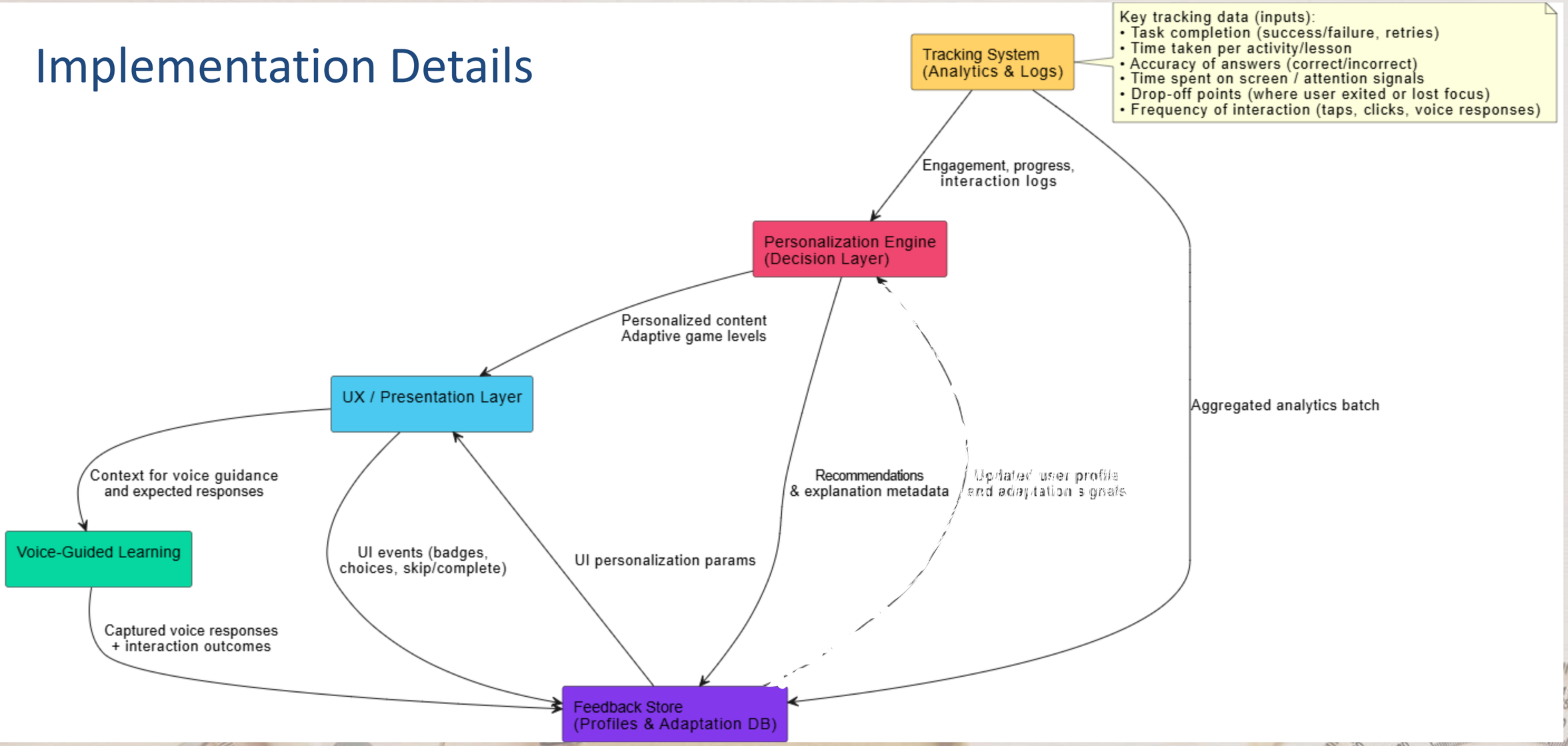
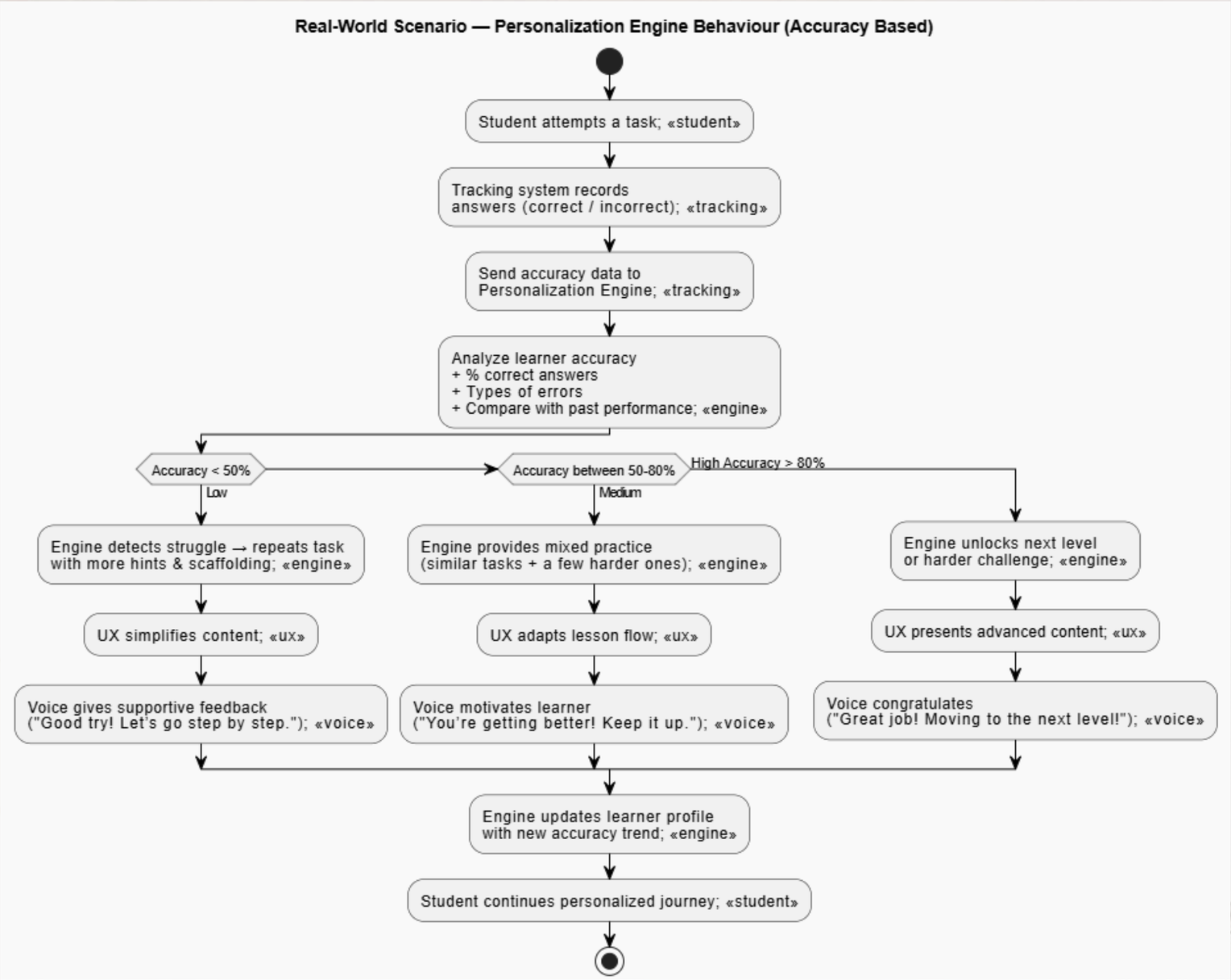
Personalization Engine

Implementation Details



Personalization Engine



IT22369574 - M.W.A.D.Wijesuriya

Specialization : Information Technology

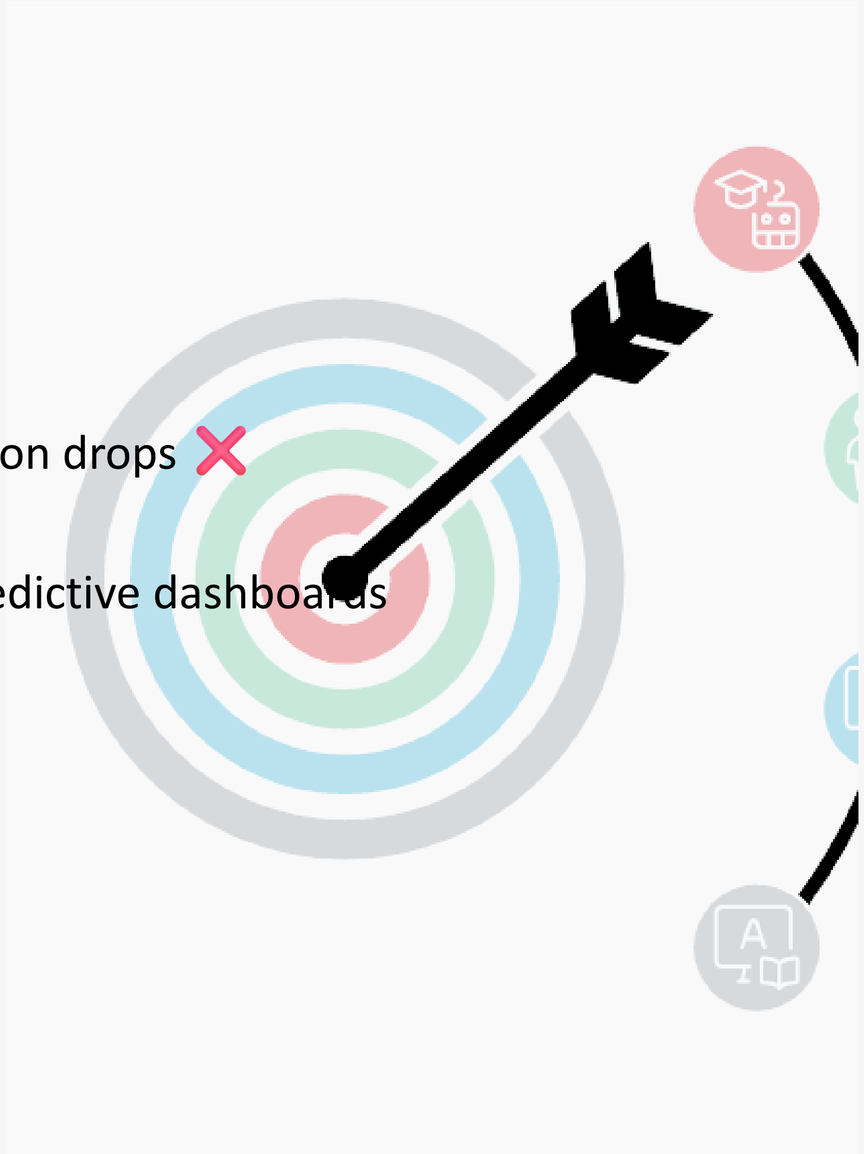
Research Question

How can an be designed to **AI-driven progress tracking system not only** monitor student engagement and task completion but also predict attention drops, stress, or fatigue, thereby enabling proactive parental and teacher intervention?

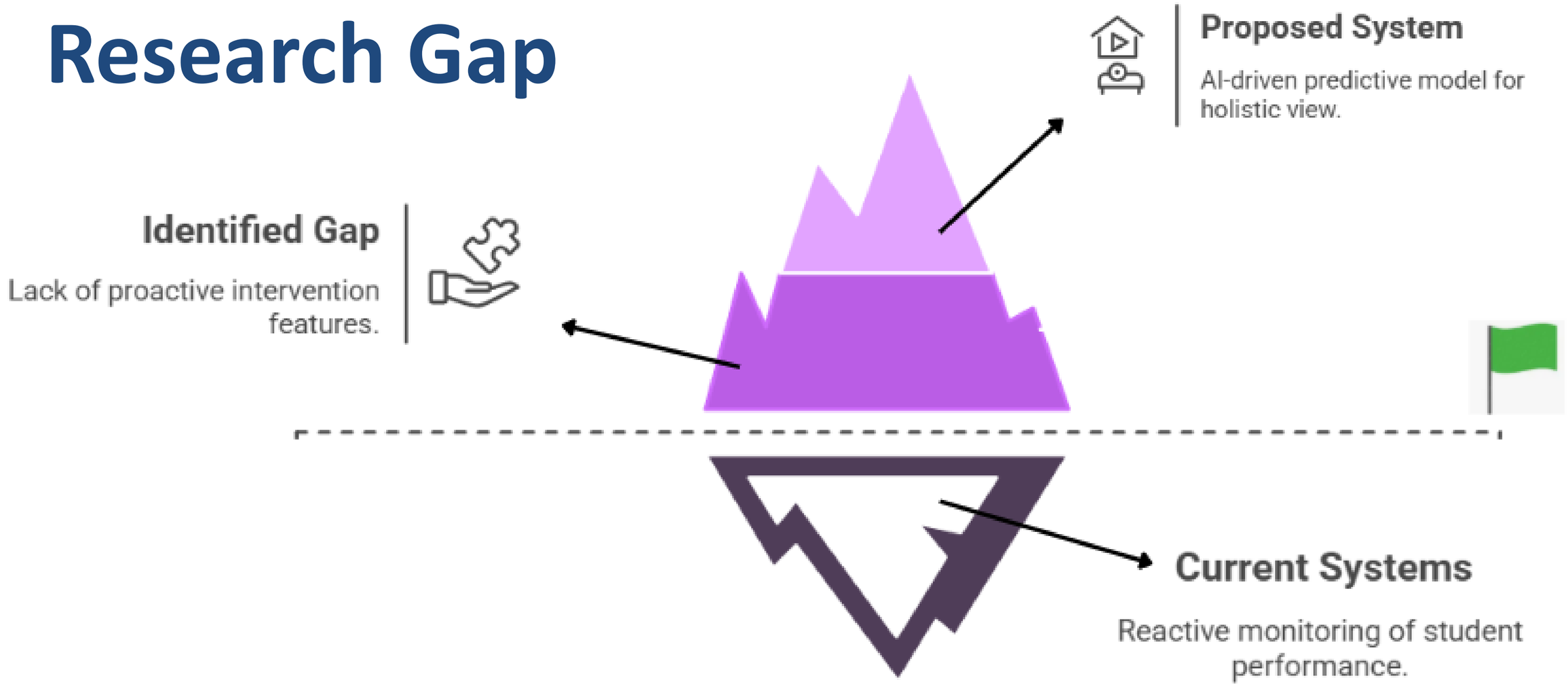
Background

Enhancing Learning with Proactive Systems

- Current progress tracking systems use passive data ❌
- Teachers and parents lack early detection tools for stress, fatigue, and attention drops ❌
- Existing LMS platforms do not offer predictive analytics ❌
- Technologies like TensorFlow, Supabase, and MERNStack enable real-time predictive dashboards ✅
- Pilot testing with MJF Charitable Foundation ensures practical validation ✅



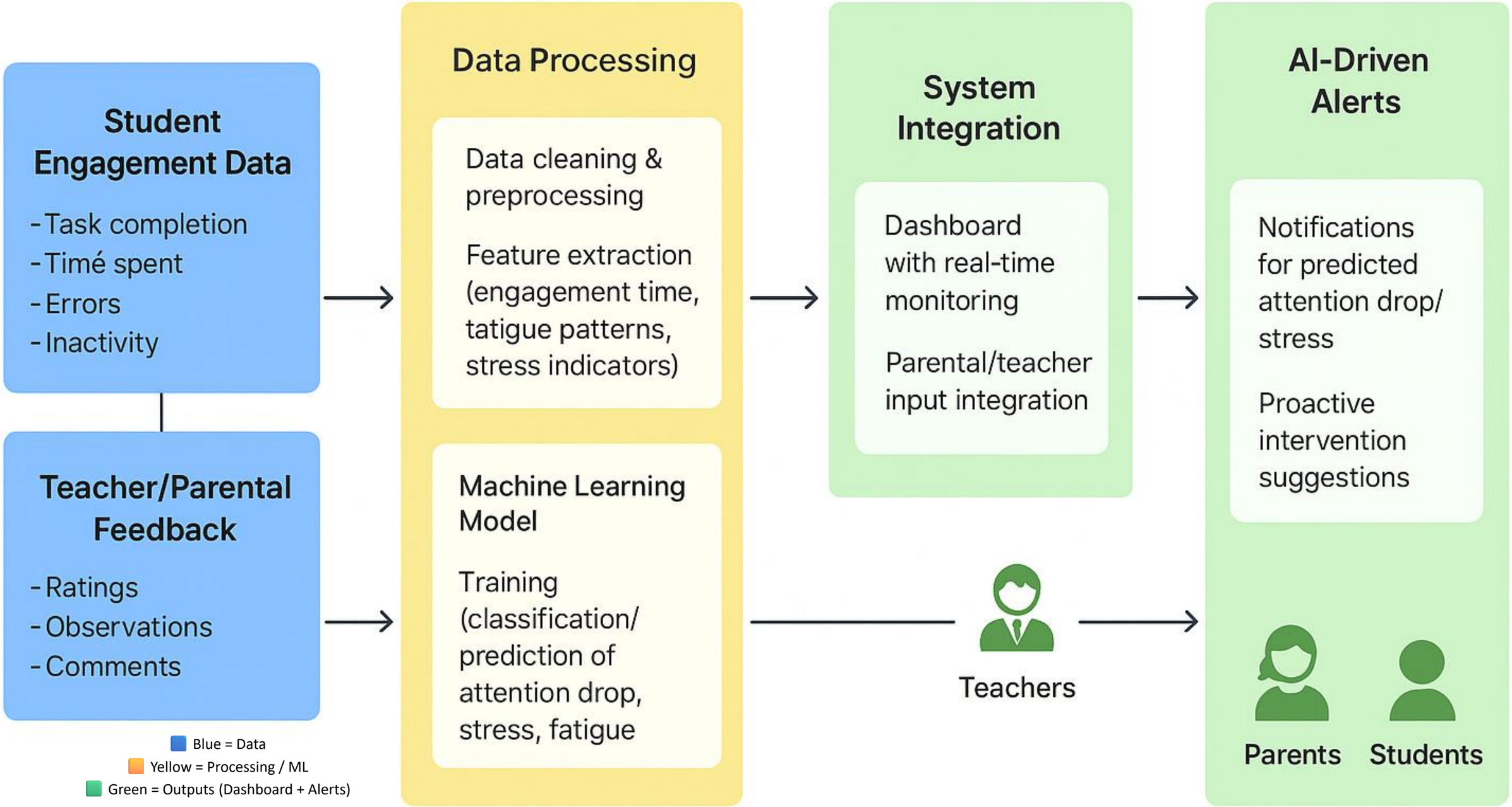
Research Gap



Unveiling the Hidden Factors Affecting Student Performance.

Research Gap

Aspect	Existing Systems	Proposed System (Your Research)	Research Gap / Novelty
Monitoring Type	❌ Passive monitoring only	✅ Predictive monitoring with AI	Lack of predictive analytics
Data Usage	❌ Limited behavioral data analysis	✅ Multi-dimensional behavioral data	Existing systems do not analyze complex behavior
Intervention	❌ Reactive alerts	✅ AI-driven proactive intervention	Absence of real-time, proactive alerts
Parental Feedback	❌ Basic, manual feedback	✅ Integrated real-time feedback system	Limited or no AI-assisted feedback
Machine Learning Utilization	❌ Limited or no ML use	✅ ML models for predicting attention & fatigue	Few leverage ML for behavior prediction
Personalization	❌ Generalized tracking	✅ Adaptive, individualized model	Lack of personalized behavioral prediction
System Alignment	❌ Focus on tracking/reporting only	✅ Therapy-aligned predictive tracking	Rare integration of mental health support


Student Engagement Data
- Task completion
- Timé spent
- Errors
- Inactivity
Teacher/Parental Feedback
- Ratings
- Observations
- Comments
Data Processing
Data cleaning & preprocessing
Feature extraction (engagement time, tatigue patterns, stress indicators)
Machine Learning Model
Training (classification/ prediction of attention drop, stress, fatigue
System Integration
Dashboard with real-time monitoring
Parental/teacher input integration
Teachers
AI-Driven Alerts
Notifications for predicted attention drop/ stress
Proactive intervention suggestions
Parents
Students
Blue = Data
Yellow = Processing / ML
Green = Outputs (Dashboard + Alerts)

Collect data from the system-Use Interaction based data

- Number of attempts: How many times students try to solve a problem
- Help request: how many times use hints
- Mouse movements: identify confusion points
- Keystrokes: recording typing speed, errors, and pauses
- Navigation patterns: Sequence of pages or modules visited, time spent each section

Get data from: Gather from the Data center in Ayati Center, Ragama, which is managed by the medical faculty of the University of Kelaniya

Key Domains of specialization-based Knowledge utilized

- Artificial Intelligence & Machine Learning
- Data Science & Analytics.
- Educational Technology.
- Software Engineering (MERN + Supabase)

Labeled data: Supervised learning

Training data: TensorFlow

Specific Latest Technologies utilized

Backend AI/ML:

- TensorFlow
- scikit-learn / TensorFlow (Python microservice)

Database & Tracking: Supabase (Real-time event tracking).

Engine Orchestration: Node.js (Express)

Integration with Frontend: React Native

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Ganepola G A S S

Specialization : Information Technology

Research Questions

How can the learning experience be enhanced for neurodiverse students by adapting tasks and interface based on their mood?

The system detects students' moods using mouse، arrow، and writing pad movements، skips، and repeated wrong answers، then adapts the interface (colors، font size، layout) and recommends the next task to improve engagement and emotional regulation.

➢ Background

- Neurodiverse students face challenges in emotional regulation and sustaining engagement. quickly changes the mood.
- Current E learning platforms are static and do not respond to students' moods or behaviors.
- Negative emotions such as frustration، boredom، or anger often lead to disengagement and poor learning outcomes.
- An adaptive system that detects mood and adjusts interface (colors، font size، layout) and recommends the next task can provide a personalized، supportive learning experience.

➢ Background

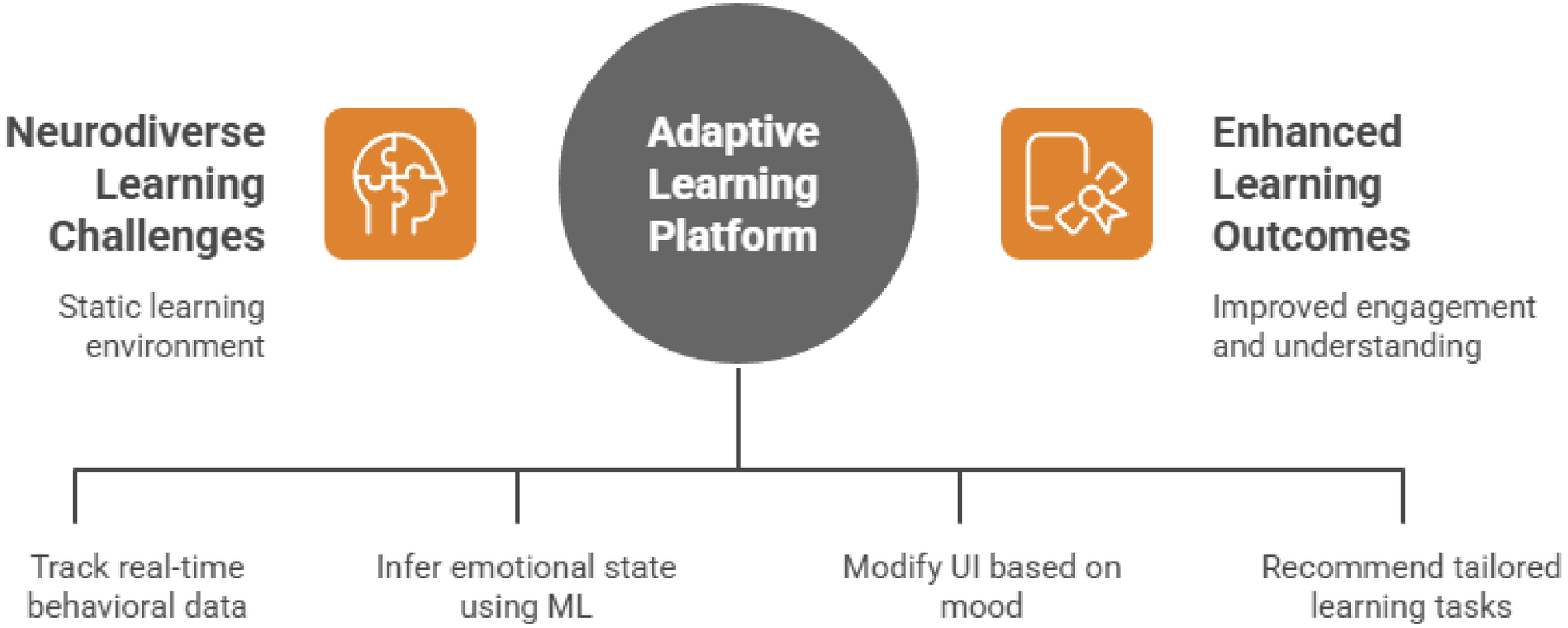
Emotional states can be inferred from behavioral signals

- Mouse and arrow movements
- Writing pad activity
- Skipping questions frequently
- Repeated wrong answers or selecting the same option

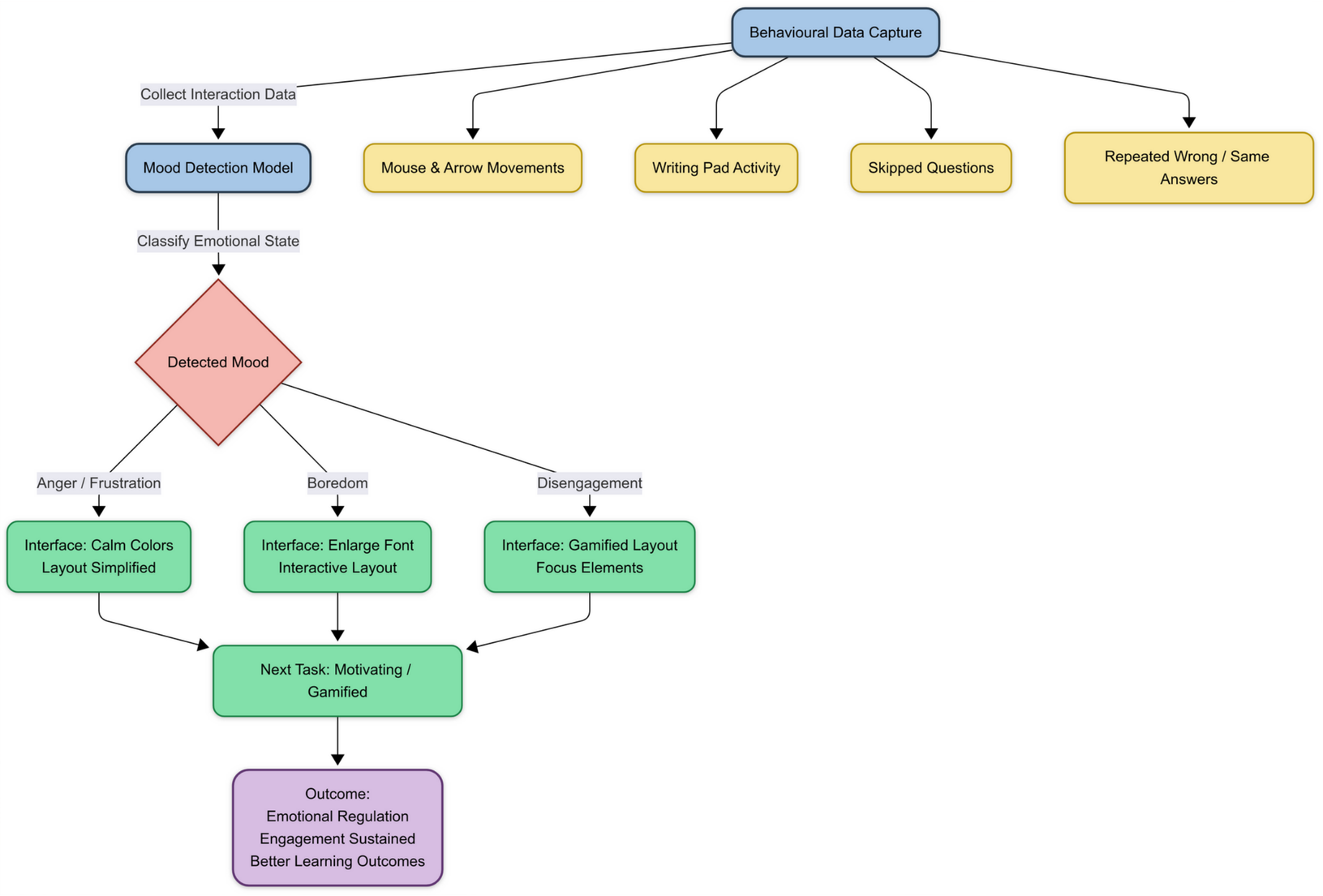
➢ Research GAP

comparison table		
Feature / System	**Existing Systems**	**Proposed System**
Adapts to mood	✗ No	✓ Yes
Behavior-based detection	✗ No	✓ Yes
Focus on neurodiverse students	✓ Yes	✓ Yes
Interface customization	✗ No	✓ Yes
Task category recommendation	✗ No	✓ Yes

ML-Driven Adaptive Learning Platform



➢ Methodology | System Diagram



Personalization Engine

Data Availability and Ethical Clearance

Data Required:

- Mouse movements, arrow key usage, writing pad strokes
- Skipped tasks and repeated wrong answers
- Mood labels for training
- Task completion time

Availability:

- Gather from the Data center in Ayati Center, Ragama which is managed by medical faculty of University of Kelaniya

Ethical Considerations:

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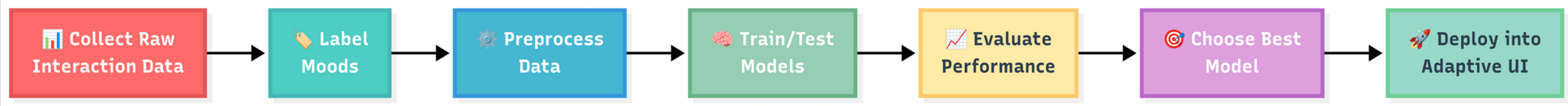

IT22926180- DE SILVA Y H E

Specialization : Information Technology

➢ Research Problem

Children with down syndrome, ASD and ADHD face challenges in speech, comprehension, and independent learning

- **Promotes independent learning**
- **Support inclusive education goals**

➢ Research Gap

Existing system do not address speech / comprehension delays in local languages

Solution

- ➢ Voice guidance in sinhala, Tamil and English
- ➢ Paired with visual reinforcement
- ➢ Enables independent interaction

➢ Research Gap Analysis

Criteria	Existing Projects
Language Support	Mostly in English; limite Sinhala/Tamil.
Focus on Neurodiverse Learners	Generalized e-learning p customization for Down
Voice-Guided Learning	Minimal or absent; so solely on text/visual l
Cultural Relevance	Use foreign content/v alignment with Sri Lar

Personalization Engine

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- TensorFlow
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➢ Commercialization

- Specialized Learning Platform - Offered to schools, therapy centers, and inclusive education institutes to support neurodiverse learners.
- Mobile & Tablet Application – Marketed to parents as an affordable home-based learning tool for children with autism, ADHD, or short attention spans.

- Growing demand for inclusive education technologies worldwide ensures a strong market.
- Schools and therapy centers are increasingly investing in digital solutions for special needs education, creating a ready customer base.
- The system's scalability allows expansion into mainstream e-learning, gamified platforms, and corporate training, where mood and engagement tracking can enhance productivity.

Thank You